

Preparing Generation Alpha for an AI-Augmented Labour Market

Strategic Insights from Sri Lanka's Education System

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Presentation Roadmap

From research problem to findings, discussion and policy implications

1	**Context**	AI-augmented labour market and Generation Alpha
2	**Study focus**	Sri Lanka’s education readiness gap
3	**Methodology**	Focus group discussion and reflexive thematic analysis
4	**Findings**	Seven barriers with participant narratives
5	**Discussion**	Comparison with past research and policy implications

Background: AI is changing future work

The study begins from the changing relationship between education, technology and employment

170M

Projected new jobs by 2030

92M

Projected displaced jobs by 2030



- AI is reshaping job structures, work tasks and required capabilities.
- Future labour markets prioritise problem-solving, adaptability and higher-order thinking.
- Education systems must prepare learners to work with, question and govern AI technologies.

Research Problem

Sri Lanka faces a preparedness gap between schooling and an AI-augmented labour market

Although AI is transforming labour markets globally, Sri Lanka's education system remains only partly aligned with the competencies required by future employment. Current reforms appear fragmented, delayed and often limited to surface-level digital exposure.



The central concern is not only the absence of devices or software. It is the lack of coordinated systemic readiness across curriculum, teaching, infrastructure, ethics and policy implementation.

Research question and objective

The inquiry focuses on barriers rather than simple technology availabilit

Research Question

What factors hinder Sri Lanka's education system in preparing Generation Alpha for an AI-augmented labour market?

Research Objective

To explore the structural, pedagogical and cultural factors that constrain the education system's ability to prepare Generation Alpha for AI-mediated work and society.

Methodology overview

Qualitative design using expert and stakeholder narratives

- Research design: qualitative inquiry.
- Data collection: focus group discussion.
- Sampling: purposive selection of policy, academic and generational perspectives.
- Analytic approach: inductive, latent-level Reflexive Thematic Analysis.



The design was appropriate because the study aimed to interpret systemic readiness, reform priorities and barriers rather than measure individual student performance.

Participant Composition

The focus group combined policy-level, academic and learner-oriented perspectives

Code	Role / position
P1	Founder of Uva Wellassa University
P2	Head of Department of Science, Faculty of Education, University of Colombo
P3	Director of Education, Ministry of Education
P4	Director, Ministry of Digital Economy
P5	Generation Z individual / student perspective

This composition helped capture the issue as a system-level readiness challenge rather than a single classroom or technology problem.

Data analysis: Reflexive Thematic Analysis

Themes were generated inductively from focus group narratives

1 **Data familiarisation**

2 **Systematic coding**

3 **Generating initial themes**

4 **Reviewing themes**

5 **Defining themes**

6 **Writing analysis**

The analysis moved beyond surface statements to interpret broader structural meanings embedded in participant narratives.

Findings overview

Seven findings show why readiness is systemic, not merely technological

Finding	Core barrier
1. Lag in AI Adoption	Delayed policy response and reactive reform
2. Shallow Digital Literacy	Tool use without deeper understanding
3. Teacher Deficit	Insufficient teacher readiness and local training capacity
4. Digital Divides	Unequal access to connectivity, devices and support
5. Curriculum Misalignment	Exam-oriented learning not aligned with AI skills
6. Fragmented Knowledge	Subjects taught in silos; weak interdisciplinary learning
7. Lack of Ethical Guidance	Absence of responsible AI literacy and governance

Finding 1: Lag in AI Adoption

Policy reacts late instead of forecasting technological change

The first finding shows that AI adoption in education is delayed and reactive. Participants framed reform as arriving after technologies have already become normal in society. Teacher training planned for the future was described as already too narrow because it treats AI mainly as ChatGPT rather than as a wider technological and labour-market transformation.

Interpretive insight

This represents policy inertia. When education systems respond after technological change has already occurred, learners are prepared for an older labour market rather than the future they will enter.

Finding 1: Narrative Evidence

Lag in AI Adoption - participant statements and analytic meaning

Participant narrative	Analytic meaning
"AI training for teachers is planned for 2026… and it is only about ChatGPT."	Training is delayed and conceptually narrow.
"We are not forecasting future trends. We act after the whole story has happened."	Policy response is retrospective rather than anticipatory.
"Misunderstanding AI as only ChatGPT is the next disaster."	AI is being reduced to one tool rather than understood as a system.

Finding 2: Shallow Digital Literacy

Using digital tools does not equal understanding AI

- The second finding shows that digital literacy remains superficial. Students may use devices and online tools, but participants argued that they are not being taught to question data, understand model behaviour, evaluate outputs, or recognise algorithmic influence.

Interpretive insight

- **AI readiness requires cognitive and critical literacy, not only operational skill. Without deeper literacy, students may become frequent technology users but weak evaluators of AI-generated information.**

Finding 2: Narrative Evidence

Shallow Digital Literacy - participant statements and analytic meaning

Participant narrative	Analytic meaning
"They learn to use tools, but not to understand how they work."	Tool operation is mistaken for competence.
"Digital literacy remains superficial - no questioning of data or models."	Students lack critical and analytical engagement.
"There is minimal exposure to ethical AI."	Digital learning excludes responsible AI awareness.

Finding 3: Teacher Deficit

Teachers lack the confidence, training and support needed for AI-enabled education

The third finding identifies teacher readiness as a major bottleneck. Participants described uneven teacher confidence, insufficient local AI training capacity, and the weakness of one-off workshops. Teacher preparedness is therefore not only a technical issue but a professional development and institutional support issue.



Interpretive insight

If teachers are not continuously supported, technology integration remains symbolic. Students may encounter devices but not meaningful AI-guided learning.

Finding 3: Narrative Evidence

Teacher Deficit - participant statements and analytic meaning

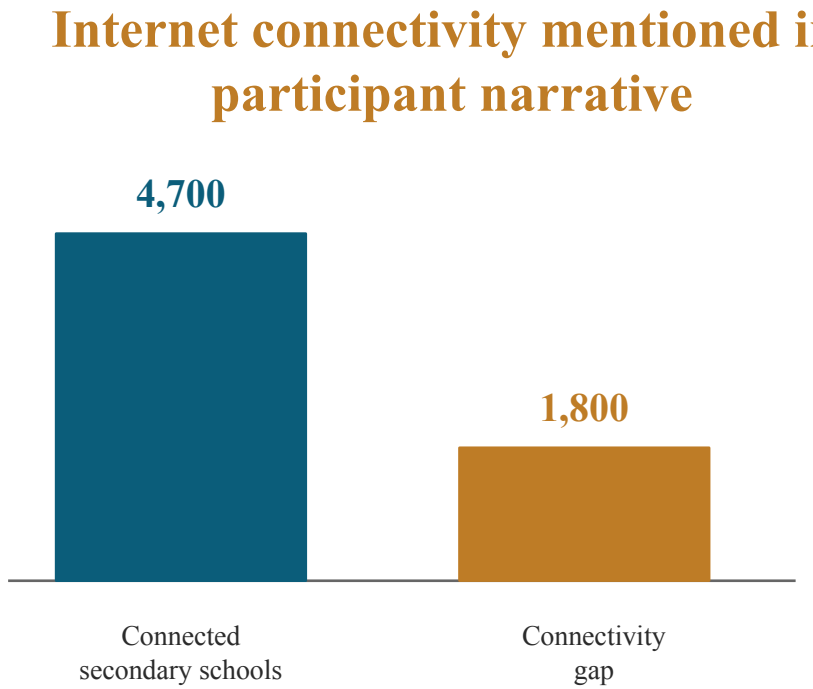
Participant narrative	Analytic meaning
"Some teachers don't like to touch computers."	Some educators still lack confidence with basic digital practice.
"Workshops are not enough - teachers need continuous support."	One-time training cannot produce sustained pedagogical change.
"We had to bring in volunteers from the USA to train teachers."	There is a shortage of local AI training capacity.

Finding 4: Digital Divides

Digital inequality affects access, skill formation and AI readiness

The fourth finding shows that AI readiness is unevenly distributed. Participants described large differences between schools with connectivity and devices and schools that lack basic infrastructure, repairs, electricity or computers. The divide is therefore not only physical access, but also maintenance, support and learning quality.

Evidence from the focus group indicated that 4,700 out of 6,500 secondary schools had internet connectivity, leaving around 1,800 schools still needing connectivity support.



Finding 4: Narrative Evidence

Digital Divides - participant statements and analytic meaning

Participant narrative	Analytic meaning
“The digital divide is very very very big.”	Inequality is perceived as severe and urgent.
“Out of 6,500 secondary schools, 4,700 schools have internet connectivity.”	Infrastructure access remains incomplete.
“Some schools have no computers at all.”	Basic device access remains a barrier.
“Many schools cannot even afford electricity or repairs for broken devices.”	The divide includes maintenance and sustainability, not only connectivity.

Finding 5: Curriculum Misalignment

The curriculum prepares students for exams more than AI-mediated worl

The fifth finding shows a mismatch between curriculum design and future labour-market skills. Participants noted the absence or marginal status of AI, machine learning and big data in formal schooling and assessment. The curriculum continues to reward recall more than real-world problem solving.

Interpretive insight

- **Students may become exam-ready but not future-ready. Meaningful reform requires changing the logic of learning, not simply inserting one AI lesson into an existing exam-oriented system.**

Finding 5: Narrative Evidence

Curriculum Misalignment - participant statements and analytic meaning

Participant narrative	Analytic meaning
"There's no AI subject in the school curriculum."	AI is not yet structurally embedded as a school-level competence.
"No plan to include AI, machine learning, or big data in A/L exams."	Assessment does not incentivize AI-related learning.
"Our assessments only evaluate knowledge - not critical thinking."	The exam system constrains innovation and application.

Finding 6: Fragmented Knowledge

Subjects are taught in silos rather than through integrated real-world problems

The sixth finding shows that knowledge remains fragmented across subject boundaries. Participants argued that AI-related problem solving requires students to combine computing, science, economics, design, environment and civic knowledge, but schooling rarely creates this kind of interdisciplinary learning experience.

Interpretive insight

- **The curriculum treats disciplines as final destinations rather than tools for solving complex problems. This weakens innovation capacity and adaptive reasoning.**

Finding 6: Narrative Evidence

Fragmented Knowledge - participant statements and analytic meaning

Participant narrative	Analytic meaning
"Subjects like physics, biology, and computing run parallel and rarely intersect." "We are educating in silos."	Knowledge is organised in isolated disciplinary tracks. The system normalises separation rather than integration.
"Real-world problems like digital finance or climate resilience are not addressed as integrated modules."	Students lack transdisciplinary problem-solving practice.

Finding 7: Lack of Ethical Guidance

AI literacy is incomplete without responsibility, privacy and accountability

The seventh finding identifies a major gap in ethical AI education. Participants expressed concern that students and teachers are not guided by national-level principles for responsible AI use. Issues such as bias, privacy, fairness, monitoring and accountability remain weakly integrated into education.



Interpretive insight

- **Technical AI competence without ethical literacy can create socially harmful outcomes. Responsible AI education must prepare learners to question how AI affects people, institutions and society.**

Finding 7: Narrative Evidence

Lack of Ethical Guidance - participant statements and analytic meaning

Participant narrative	Analytic meaning
"There is no national guideline on ethical AI use in education."	The system lacks formal responsible-AI direction.
"Who will control and monitor how students use these tools?"	Governance responsibility remains unclear.
"Ethics was one of the top three problems identified."	Ethics is seen as a central readiness issue, not a secondary topic.

Cross-finding synthesis

The seven findings form a cycle of unpreparedness

- Delayed policy response slows teacher preparation and curriculum reform.
- Teacher deficit and shallow digital literacy reinforce one another in the classroom.
- Digital divides make AI readiness uneven across schools and regions.
- Curriculum misalignment and fragmented knowledge prevent real-world AI problem solving.
- Weak ethical guidance leaves students technically exposed but socially underprepared.

Overall, AI readiness is not a single technology issue. It is a system condition shaped by policy, teachers, infrastructure, curriculum, knowledge integration and ethics.

Discussion I: AI adoption and digital literacy

Comparison with past research: tool exposure is not enough

Past research	This study's finding	Comparison
UNESCO (2021) and OECD (2021) argue that education systems require anticipatory AI governance.	Sri Lanka's education response appears delayed and reactive.	The finding supports the need for forward-looking policy rather than late adoption.
Selwyn (2019) warns that technology use is often mistaken for real digital competence.	Students use tools but rarely question data, models or AI outputs.	The study confirms this distinction in the Sri Lankan context.
Holmes, Bialik and Fadel (2019) emphasize critical AI literacy and understanding of algorithmic systems.	Digital literacy is shallow and operational rather than analytical.	The finding extends global arguments to the issue of Generation Alpha readiness.

Discussion II: Teachers and digital inequality

Comparison with past research: readiness depends on support structures

Past research	This study's finding	Comparison
Mishra and Koehler (2006) explain that effective technology use requires technological, pedagogical and content knowledge.	Teachers lack confidence and continuous AI-related support.	The finding shows that teacher readiness must include pedagogy, not only device operation.
Ertmer and Ottenbreit-Leftwich (2010) argue that teacher change depends on knowledge, confidence, beliefs and institutional culture.	One-off workshops are viewed as insufficient.	The finding supports continuous professional development rather than isolated training.
van Dijk (2020) argues that digital inequality includes access, skills, use and outcomes.	Some schools lack computers, repairs, electricity or connectivity.	The study confirms digital divide as multidimensional, not merely connectivity-based.

Discussion III: Curriculum, interdisciplinarity and ethics

Comparison with past research: future readiness requires deeper reform

Past research	This study's finding	Comparison
World Economic Forum (2023) highlights growing demand for analytical thinking, adaptability and complex problem-solving.	The curriculum remains exam-focused and recall-oriented.	The finding reveals a mismatch between school outputs and labor-market capabilities.
OECD (2021) and UNESCO (2021) emphasize interdisciplinary and application-oriented learning for AI futures.	Subjects are taught in silos with few integrated real-world modules.	The study shows that fragmentation limits problem-solving for complex societal challenges.
OECD (2021) and UNESCO (2021) stress privacy, bias, fairness and accountability in AI education.	There is no clear national guidance on ethical AI use in education.	The finding shows that AI readiness must include responsible citizenship, not only technical skill.

Overall discussion: What the comparison shows

The paper adds Sri Lankan qualitative evidence to global AI-readiness debates

- Past studies show that AI readiness requires digital literacy, teacher capacity, equitable access, curriculum reform and ethical governance.
- This study finds the same dimensions in Sri Lanka but shows that they are interconnected and institutionally embedded.
- The main contribution is the qualitative explanation of how policy delay, exam culture and digital inequality combine to weaken Generation Alpha's readiness.
- Therefore, AI readiness should be treated as educational transformation, not as a narrow technology adoption programme.



Recommendations

Three reforms proposed by the study

1 **Embed AI literacy with ethics**

Introduce AI concepts, critical data awareness, privacy, bias and accountability into school learning.

2 **Strengthen teacher development**

Move from one-off workshops to continuous training, mentoring and local AI trainer capacity.

3 **Pilot AI-ready learning ecosystems**

Combine curriculum reform, infrastructure, interdisciplinary projects and equitable access.

Conclusion

Sri Lanka is not yet fully AI-ready, but reform pathways are clear

The findings show that Sri Lanka's education system faces significant structural, pedagogical and cultural barriers to preparing Generation Alpha for an AI-augmented labour market.

- The issue is not only lack of technology; it is lack of coordinated readiness.
- AI literacy must include critical understanding and ethical responsibility.
- Teacher development, curriculum reform and equitable infrastructure must progress together.
- Preparing Generation Alpha requires system-wide educational transformation.

Key references used in discussion

Selected references from the paper

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Thank you

Sri Lanka’s Education System and Generation Alpha: Ready for the AI-Augmented Labour Market?

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